# MONTHLY RETAIL SALES

## Project Report

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**Role** : Aspiring Data Analyst

**Tool Used** : Microsoft Excel

**Project Type** : Data Cleaning, Data Analysis, and Dashboard Development

**Project Focus on** : Data cleaning, Data Analysis Dashboard development

**Project Duration** : 2 Days

# 1.Project Overview

## 1.1 Overview

The Monthly Retail Sales Analysis and Interactive Dashboard using Microsoft Excel project focuses on cleaning, organizing, analyzing, and visualizing retail sales data to understand business performance and support better decision-making. Retail businesses generate large amounts of data related to products, customers, sales transactions, discounts, costs, profits, and payment methods. Analy zing this data helps identify business trends, evaluate performance, and discover areas that may require improvement.
The project was developed using Microsoft Excel and its data analysis and visualization features, including Excel formulas, PivotTables, Pivot Charts, GETPIVOT DATA (), slicers, and dashboard design techniques. The main objective was to transform raw retail sales data into a structured and interactive dashboard that presents important business information in an understandable visual format.

## 1.2 Dataset Preparation and Data Cleaning

The initial dataset contained 10,031 rows. Before performing the analysis, data cleaning was carried out to remove duplicate records and rows containing blank values in selected fields. This process was intended to improve data consistency and prepare the dataset for analysis.

**The cleaning process recorded the following changes:**

- Initial dataset: 10,031 rows
- After removing duplicates: 10,001 rows
- After deleting rows with blank names: 9,881 rows
- After deleting rows with blank cities: 9,801 rows
- After deleting rows with blank discounts: 9,761 rows
- After deleting rows with blank ratings: 9,721 rows
- After deleting rows with blank payment methods: 9,663 rows

Following these steps, the working dataset contained 9,663 rows, according to the recorded cleaning results. These steps helped prepare the data for the next stage of analysis. The final cleaned records were used as the basis for creating summaries and visualizations.

## 1.3 Data Analysis and Dashboard Development

After data cleaning, the project focused on analyzing sales and profit performance from different business perspectives. PivotTables were used to summarize the data, while Pivot Charts converted the summaries into visual representations. GETPIVOT DATA () was used to retrieve values from PivotTables where required, and slicers were incorporated to provide interactive filtering.

**The dashboard includes eight major visualizations:**

1. **Sales by Category:** A clustered bar chart used to compare sales across product categories and identify categories that contribute strongly to revenue
2. **Profit by State:** A clustered bar chart used to compare profit across states and identify stronger-performing geographic markets.
3. **Monthly Sales Trend:** A line chart designed to show how sales change over time and help identify monthly patterns.
4. **Sales by Customer Segment:** A doughnut or bar chart used to compare the sales contribution of different customer segments.
5. **Top 10 Products by Profit:** A clustered bar chart used to highlight the ten products with the highest total profit and support product-level performance analysis.
6. **Sales by Salesperson:** A clustered bar chart used to compare sales performance across salespeople and identify stronger contributors to sales.
7. **Order Status Distribution:** A doughnut chart used to display the proportion of orders across different statuses and provide an overview of order processing.

8. **Sales by Payment Method:** A clustered chart used to compare the use of different payment methods and understand customer payment preferences.

## 1.4 Purpose and Expected Outcome

- To provide a clear overview of retail sales performance through a single Excel dashboard.
- To clean raw data by handling duplicate records, blank values, and data quality issues.
- To analyze sales, profit, product categories, states, and salesperson performance using PivotTables.
- To create charts that present sales trends, profitable products, order statuses, and payment methods.
- To use interactive slicers to filter data and explore different aspects of business performance.
- To convert raw retail data into meaningful insights and develop practical skills in data analysis and dashboard development using Microsoft Excel.

# 2.Problem Statement

Retail sales data contains important information about sales, profit, products, customers, states, and payment methods. However, analyzing large amounts of raw data manually can be time-consuming and make it difficult to understand overall business performance. Duplicate records, blank values, and data quality issues may also affect the accuracy of the analysis.

Without proper analysis, it can be difficult to identify which product categories generate the highest sales, which states contribute the most profit, which products are most profitable, and how salespeople perform. It is also challenging to understand monthly sales trends and the distribution of order statuses and payment methods.

Therefore, this project focuses on cleaning, organizing, and analyzing retail sales data using Microsoft Excel. PivotTables, Pivot Charts, KPI cards, and interactive slicers are used to summarize the data and present important information in a single dashboard. The dashboard helps users compare business performance, identify useful patterns, and understand retail sales results more efficiently.

# 3.Objectives

- I aim to **evaluate** overall retail performance by analyzing total sales, total profit, total cost, total quantity, and average customer rating.
- I want to **identify** the highest-selling product category and the state generating the highest profit.
- I aim to **analyze** monthly sales trends, identify the top 10 products by profit, and compare salesperson performance.
- I want to **examine** sales distribution across customer segments, order statuses, and payment methods.
- I use interactive month slicers to **filter and compare** KPIs and sales performance across selected months.
- Through this project, I aim to **develop** an interactive Excel dashboard that presents meaningful business insights and supports better retail decision-making.

# 4.Tools Used

I used the following tools and features in Microsoft Excel 2024 to complete my retail sales analysis and dashboard project.

1. **Microsoft Excel 2024** – For data processing, analysis, calculations, and dashboard creation.
2. **Data Cleaning** – For handling duplicate records, blank values, and data quality issues.
3. **Excel Formulas and Functions** – For calculating totals, averages, and key performance indicators (KPIs).
4. **PivotTables** – For summarizing and analyzing sales, profit, products, and other business data.
5. **Pivot Charts** – For visualizing sales trends, profit comparisons, and business performance.
6. **Slicers** – For interactively filtering dashboard data by month.
7. **Excel Shapes and Text Boxes** – For designing KPI cards and arranging the dashboard layout.
8. **GETPIVOTDATA Function** – For retrieving summarized values from PivotTables to display in KPI cards.
9. **Chart Formatting and Dashboard Design** – For improving the dashboard's appearance, readability, and presentation.

# 5.Dataset Description

## 5.1 Dataset Overview

I used a retail sales dataset containing 20 columns that provide information about orders, customers, products, sales transactions, costs, profits, and payment details. The dataset helps me analyze different aspects of retail business performance and understand how sales vary across product categories, locations, customer segments, and salespeople. I used Microsoft Excel 2024 to clean, organize, and analyze the data and prepare an interactive dashboard.

## 5.2 Order Information

- **Order_ID:** This column contains a unique identifier for each order, helping me identify and track individual transactions.
- **Order_Date:** This column records the date of each order. I used this information to analyze monthly sales trends and compare performance across different periods.

## 5.3 Customer Information

- **Customer_Name:** This column contains customer names and helps identify customer records.
- **Customer_Age:** This column shows the age of customers. It can help me explore customer age patterns and understand the age groups represented in the dataset.
- **Customer_Segment:** This column categorizes customers into different segments, allowing me to compare sales across customer groups.

## 5.4 Geographical Information

- **City:** This column identifies the city associated with an order and helps me examine sales across different cities.

- **State:** This column identifies the state associated with an order. I used it to compare profit across states and identify which states contribute the most profit.

## 5.5 Product Information

- **Category:** This column groups products into categories, such as Clothing, Electronics, Furniture, Grocery, and Office Supplies. It helps me compare sales performance across product categories.
- **Product:** This column contains the names of individual products. I used it to identify the top 10 products based on profit.

## 5.6 Sales and Quantity Information

- **Quantity:** This column records the number of units purchased in an order. It helps me understand product demand and calculate total quantity sold.
- **Unit_Price:** This column shows the price per unit of a product and provides information about product pricing.
- **Discount:** This column records the discount applied to an order. It helps me examine discount information alongside sales and profit.

## 5.7 Financial Information

- **Sales:** This column represents the sales amount associated with an order. I used it to calculate total sales, compare categories, and analyze monthly sales performance.
- **Cost:** This column records the cost associated with an order. It helps me understand the costs involved in retail transactions.
- **Profit:** This column represents the profit associated with an order. I used it to compare profitability across states, products, and salespeople.

## 5.8 Customer Feedback

- **Rating:** This column contains customer ratings. I used it to calculate the average customer rating and include it as a key performance indicator (KPI) in the dashboard.

## 5.9 Payment and Sales Information

- **Payment_Method:** This column identifies the payment method used for an order. It helps me analyze how sales are distributed across different payment methods.
- **Sales_Channel:** This column identifies the channel through which a sale was made. It can help me compare sales across different channels.
- **Order_Status:** This column shows the status of an order. I used it to create an order status distribution chart and understand the distribution of order statuses.
- **Salesperson:** This column identifies the salesperson associated with an order. I used it to compare sales performance across salespeople.

## 5.10 Dataset Preparation and Analysis

Before creating the dashboard, I cleaned and prepared the dataset in Microsoft Excel 2024. During the cleaning process, I removed duplicate records and handled blank values in relevant columns, including customer names, cities, discounts, ratings, and payment methods. I then used Excel formulas, PivotTables, Pivot Charts, and interactive month slicers to analyze the data and present the results visually.

The prepared dataset supports the development of an interactive retail sales dashboard featuring five key performance indicators (KPIs) and eight charts. These visualizations help me examine

sales performance, profitability, product performance, customer segments, and other important business metrics in one place.

# 6. Data Cleaning

Data cleaning was an important step in my Monthly Retail Sales Analysis project. I used Microsoft Excel 2024 to prepare the raw dataset by removing duplicate records, handling blank values, correcting inconsistent categories, standardizing dates, and cleaning customer ratings. This process helped me prepare the data for accurate analysis and dashboard development.

## 6.1 Removing Duplicate Records

Initially, my dataset contained **10,031 rows**. I used Excel's **Remove Duplicates** feature to remove duplicate records and reduce the dataset to **10,001 rows**. This helped avoid counting duplicate transactions during analysis.

## 6.2 Handling Blank Values

I checked important columns for blank values and removed records containing blanks in the following fields. The row counts after each step were:

| Cleaning Step | Remaining Rows |
|---|---|
| Initial dataset | 10,031 |
| After removing duplicates | 10,001 |
| After deleting blank customer names | 9,881 |
| After deleting blank cities | 9,801 |
| After deleting blank discounts | 9,761 |
| After deleting blank ratings | 9,721 |
| After deleting blank payment methods | 9,663 |

After completing these steps, the dataset contained **9,663 rows**.

## 6.3 Formulas and Excel Features Used

I used the following Excel formulas and features to support the data-cleaning process.

- **TRIM:** Removed unnecessary spaces from text values.
- **CLEAN:** Removed non-printing characters from text values. I combined it with TRIM to improve text consistency.
- **PROPER:** Standardized the capitalization of names and text values.

**=PROPER(TRIM(CLEAN(A2)))**

- **COUNTIF:** Helped identify repeated values in a column.

**=COUNTIF(A:A,A2)**

Values greater than 1 indicate that the value appears multiple times; however, this alone does not prove that an entire order record is a duplicate.

- **IFERROR and DATEVALUE:** Helped convert date text into valid Excel dates and handle conversion errors.

**=IFERROR(DATEVALUE(B2),"Check Date")**

- **Find and Replace:** Helped correct inconsistent category names, city names, and other text values, such as replacing abbreviated city names with their standardized names.
- **Remove Duplicates:** Used the built-in Excel feature to remove duplicate records.
- **Filter:** Used column filters to locate blank cells and inspect inconsistent values before cleaning.

## 6.4 Standardizing Dates and Ratings

I checked the order dates because they appeared in different formats, such as 24/05/2025 and 24-05-2025. I worked on standardizing the dates so that Excel could recognize them consistently for monthly analysis and PivotTables.

I also checked the Rating column for inconsistent values and formatting. I aimed to display ratings consistently, such as 3.4, instead of showing unwanted decimal places or totals.

## 6.5 Final Result

After removing duplicate records and handling blanks in the selected columns, I reduced the dataset from **10,031 rows to 9,663 rows**. This prepared dataset was then used to create PivotTables, Pivot Charts, KPI cards, and interactive slicers for my retail sales dashboard.

# 7. Data Analysis

After completing the data-cleaning process, I used Microsoft Excel 2024 to analyze the retail sales dataset. I created PivotTables to summarize sales information, compare business performance, and identify useful insights. I also created KPI cards using values from the PivotTables to display important business metrics on my dashboard.

## 7.1 Creating PivotTables

- **I created PivotTables from the cleaned dataset using the following steps:**
  1. I selected the cleaned dataset, including the column headings.
  2. I clicked the **Insert** tab and selected **PivotTable**.
  3. I selected the option to place the PivotTable in a new worksheet and clicked **ok.**
  4. I dragged the required fields into the **Rows, Columns, Values, and Filters** areas based on the business question.
  5. I changed the value calculation to **Sum, Count, or Average**, depending on the type of analysis required.
  6. I repeated the process to create PivotTables for different aspects of retail performance.
- **I created the following PivotTables from the cleaned dataset:**
  1. **Sales by Category:** Category in Rows and Sales in Values.
  2. **Profit by State:** State in Rows and Profit in Values.
  3. **Monthly Sales Trend:** Order_Date grouped by month in Rows and Sales in Values.
  4. **Sales by Customer Segment:** Customer_Segment in Rows and Sales in Values.

5. **Top 10 Products by Profit:** Product in Rows and Profit in Values, sorted from largest to smallest and filtered to the top 10.
6. **Sales by Salesperson:** Salesperson in Rows and Sales in Values.
7. **Order Status Distribution:** Order_Status in Rows and Order_ID in Values, summarized using Count.
8. **Sales by Payment Method:** Payment_Method in Rows and Sales in Values.

## 7.2 Creating Key Performance Indicators (KPIs)

I created KPI cards to display important business metrics at the top of the dashboard. I used the summarized values from PivotTables rather than calculating every metric separately from the raw dataset.

### ✓ KPIs Image:




➢ **The KPIs included:**

1. **Total Sales:** The sum of sales values.
2. **Total Profit:** The sum of profit values.
3. **Total Cost:** The sum of cost values.
4. **Total Quantity:** The sum of quantities sold.
5. **Average Rating:** The average of customer ratings.

## 7.3Retrieving KPIs Values from PivotTables

I used cell reference formulas such as =Sheet!A1 to retrieve summarized values from PivotTables and display them in dashboard KPI cards. This allowed me to show important business metrics on the dashboard without entering the values manually.

- **Steps I followed to create the PivotTables:**
    1. I created a PivotTable containing the required metric, such as Sum of Sales.
    2. I selected the dashboard cell where I wanted to display the KPI value.
    3. I entered = and selected the relevant value cell in the PivotTable to create a cell reference formula, such as =Sheet!A1.
    4. I used the linked value to display the KPI in a dashboard cell or connected it to a text box or shape.
    5. I repeated the process for total profit, total cost, total quantity, and average customer rating.

## 7.4 Formatting KPI Cards

After retrieving the values, I used Excel shapes and text boxes to design KPI cards. I linked the cards to the cells containing the KPI values so that the displayed numbers could update when the underlying values changed. I formatted the numbers, added suitable titles, and arranged the cards at the top of the dashboard to make the main business metrics easy to understand.

## 7.5 Creating Charts and Interactive Slicers

I used Pivot Charts to present the summarized data visually through bar charts, column charts, line charts, and doughnut charts. I also added month slicers to filter the dashboard by selected months.

I connected the slicers to the relevant PivotTables using **Slicer → Report Connections** (or **PivotTable Connections**, depending on the Excel interface). This allowed the connected charts and KPI values to update when a month was selected.

## 7.6 Result of Data Analysis

By combining PivotTables, cell reference formulas such as =Sheet!A1, KPI cards, Pivot Charts, and interactive slicers, I developed a single Excel dashboard to analyze retail sales performance. It allows me to compare sales and profitability, identify top-performing products and categories, examine monthly sales trends, and understand other important business metrics in a clear and organized way.

# 8. Dashboard Design

I designed an interactive retail sales dashboard using Microsoft Excel 2024 to present important business information in one place. I arranged KPI cards at the top and charts below them to make sales performance easy to understand. I used charts, slicers, formulas, and Excel shapes to create a clear and user-friendly dashboard.

## 8.1 KPIs Cards

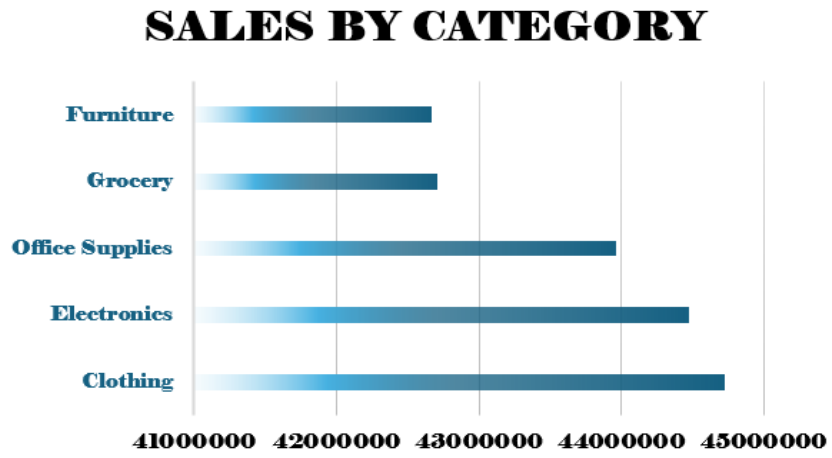



I placed KPI cards at the top of the dashboard to display five important business metrics: total sales, total profit, total cost, total quantity, and average customer rating. I linked the KPI values to cells containing the summarized PivotTable results using cell reference formulas such as =Sheet!A1. This helped me display the values without entering them manually.

## 8.2 Charts and Visualizations

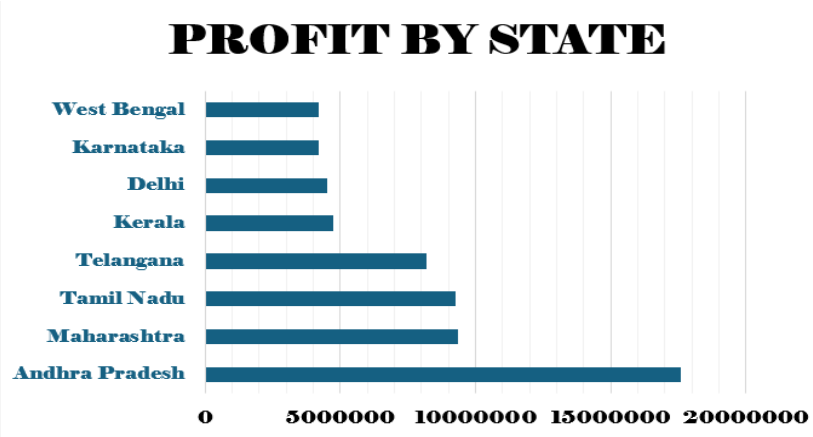
I created eight charts in Microsoft Excel 2024 to analyze different aspects of retail sales performance. I used PivotTables and Pivot Charts to summarize the data and present the results visually. Each chart focuses on a specific business question and helps me understand sales, profit, product performance, and customer-related information.

## 1. Sales by Category



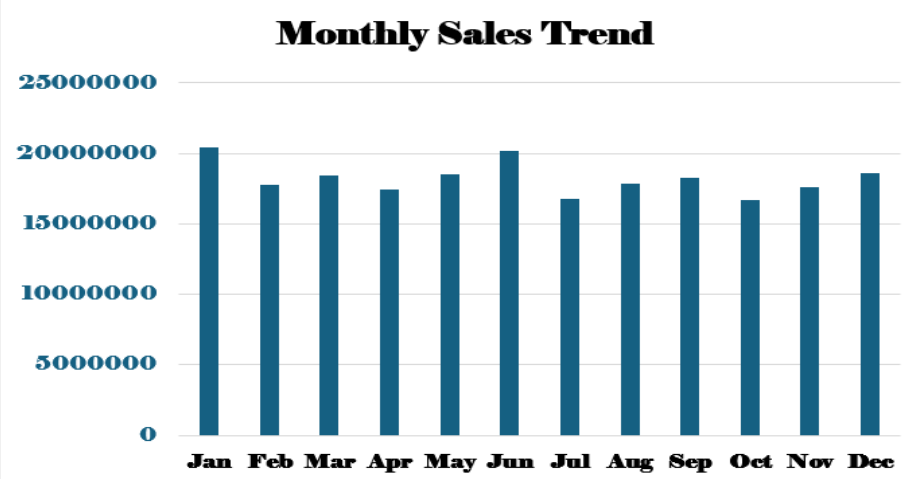

- I created a PivotTable with Category in the Rows area and Sales in the Values area, summarized by Sum.
- I inserted a clustered bar chart using the summarized PivotTable data to compare sales across Clothing, Electronics, Furniture, Grocery, and Office Supplies.
- I formatted the chart with a suitable title, axis labels, and clear presentation to make the sales comparison easy to understand.
- I used the chart to identify the product category generating the highest sales and compare the sales performance of all categories.
- This analysis helped me understand which product categories contribute more to retail sales revenue and compare their overall performance.

## 2.Profit by State



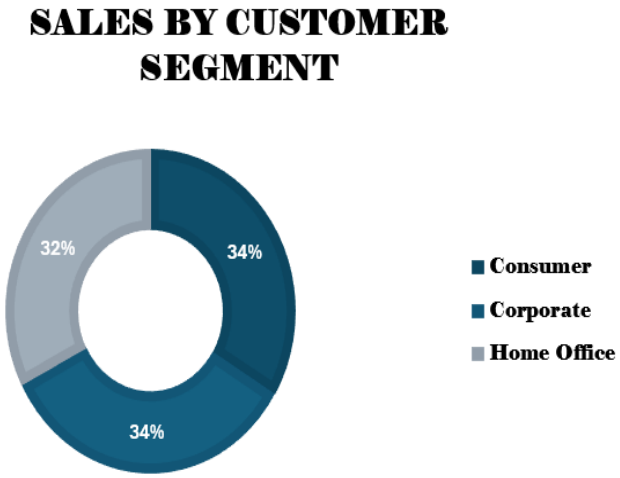

- I created a PivotTable with State in the Rows area and Profit in the Values area, summarized by Sum.
- I inserted a clustered bar chart using the summarized Pivot-Table data to compare total profit across different states.
- I formatted the chart with a suitable title and clear presentation to make the comparison easy to understand.
- I used the chart to identify the states generating the highest and lowest profit.
- This analysis helped me understand state-level business performance and identify locations that may require further investigation to improve profitability.

## 3.Monthly Sales Trend



- I created a PivotTable with Order_Date grouped by month in the Rows area and Sales in the Values area, summarized by Sum.
- I inserted a column chart using the summarized PivotTable data to compare total sales from January to December.
- I added the chart title Monthly Sales Trend and displayed the months on the horizontal axis and total sales on the vertical axis.
- I used the chart to identify months with relatively high and low sales.
- This analysis helped me compare monthly sales performance and understand changes in sales across the year.

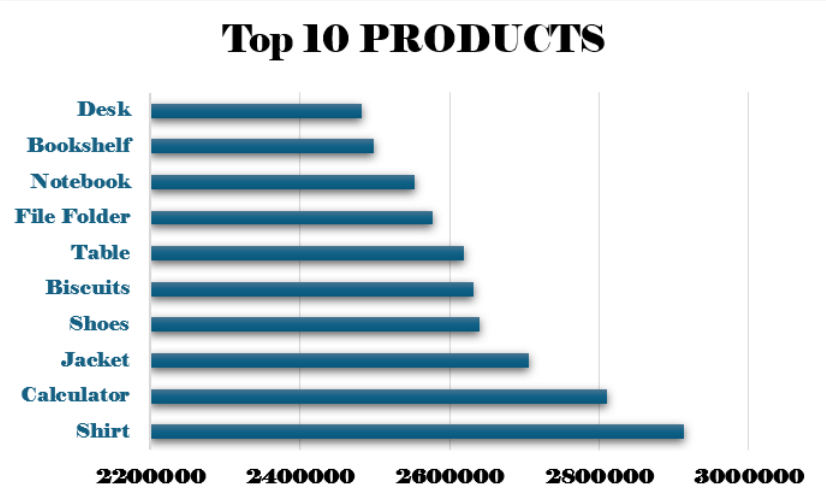
**4. Sales by Customer Segment**




- I created a PivotTable with Customer_Segment in the Rows area and Sales in the Values area, summarized by Sum.
- I inserted a doughnut using the summarized PivotTable data to compare total sales across different customer segments.
- I formatted the chart with a suitable title and clear presentation to make the comparison easy to understand.

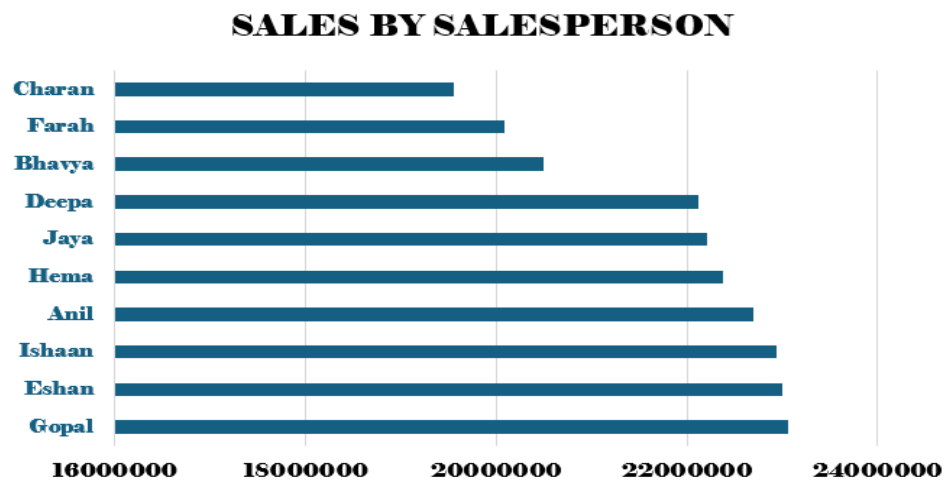
- I used the chart to identify which customer segments contribute more to total sales and compare sales across customer groups.
- This analysis helped me understand customer segment performance and how the business can plan suitable marketing activities.

## 5. Top 10 Products by Profit



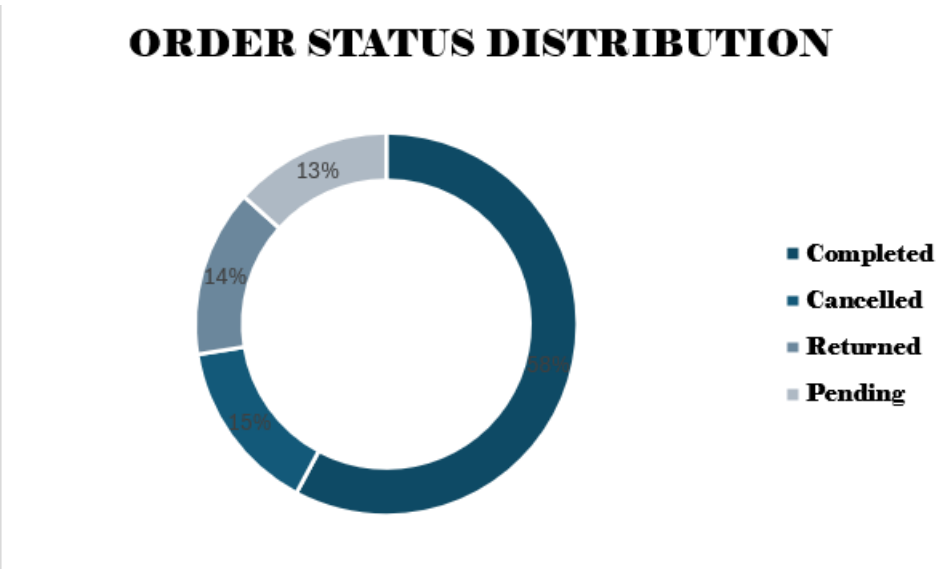

- I created a PivotTable with Product in the Rows area and Profit in the Values area, summarized by Sum.
- I sorted the profit values from largest to smallest and applied a Top 10 value filter to display the ten most profitable products.
- I inserted a Clustered bar chart using the summarized PivotTable data to compare the profits of the top 10 products.
- I formatted the chart with a suitable title and clear presentation to make the comparison easy to understand.
- I used the chart to identify the products generating the highest profit and understand which products contribute more to overall business profitability.

## 6. Sales by Salesperson



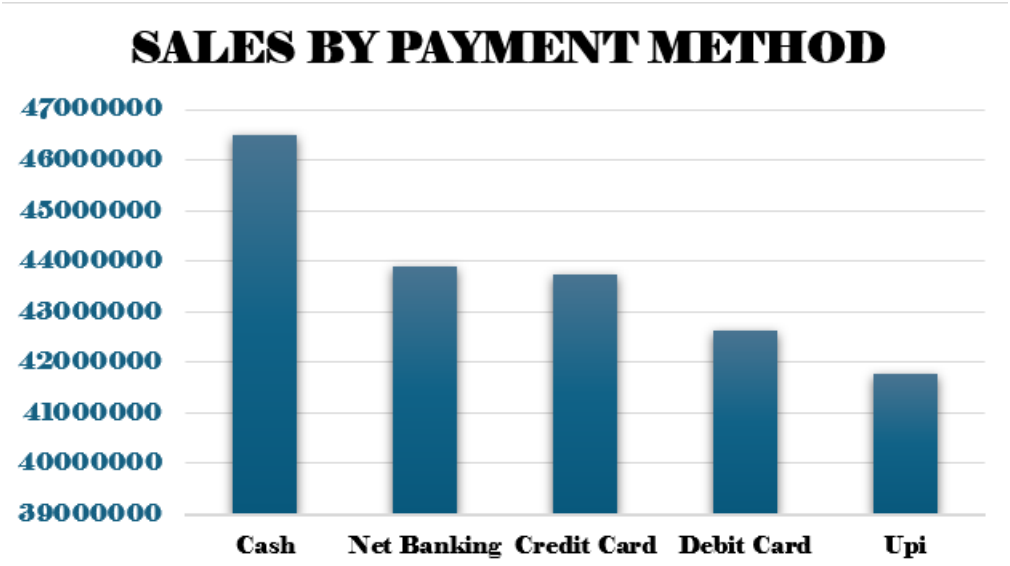

- I created a PivotTable with Salesperson in the Rows area and Sales in the Values area, summarized by Sum.
- I inserted a clustered bar chart using the summarized PivotTable data to compare total sales across different salespeople.
- I formatted the chart with a suitable title and clear presentation to make the comparison easy to understand.
- I used the chart to identify salespeople with higher and lower recorded sales.
- This analysis helped me understand differences in salesperson performance and identify areas where additional support or further analysis may be useful.

## 7. Order Status Distribution



- I created a PivotTable with Order_Status in the Rows area and Order_ID in the Values area, summarized by Count.
- I inserted a doughnut chart using the summarized PivotTable data to show the number of orders across different order statuses.
- I formatted the chart with a suitable title and clear presentation to make the order distribution easy to understand.
- I used the chart to compare the number of orders in each status, such as Completed, Pending, or Cancelled, if these values are present in my dataset.
- This analysis helped me understand order distribution and identify order statuses that may require further attention from the business.

**8. Sales by Payment Method**




- I created a PivotTable with Payment_Method in the Rows area and Sales in the Values area, summarized by Sum.
- I inserted a clustered column chart using the summarized PivotTable data to compare total sales across different payment methods.
- I formatted the chart with a suitable title and clear presentation to make the comparison easy to understand.

- I used the chart to identify which payment methods are associated with higher and lower recorded sales.
- This analysis helped me understand how sales are distributed across different payment methods and compare the sales performance of the available payment options.

### 9.Overall Result

These eight charts convert summarized retail data into visual information that is easier to understand and compare. Together, they help me analyze product categories, state profitability, monthly sales trends, customer segments, profitable products, salesperson performance, order statuses, and payment methods. When connected to the relevant PivotTables and month slicers, the charts also allow users to explore selected periods and compare retail business performance more efficiently.

## 8.3 Slicers and Interactive Filters

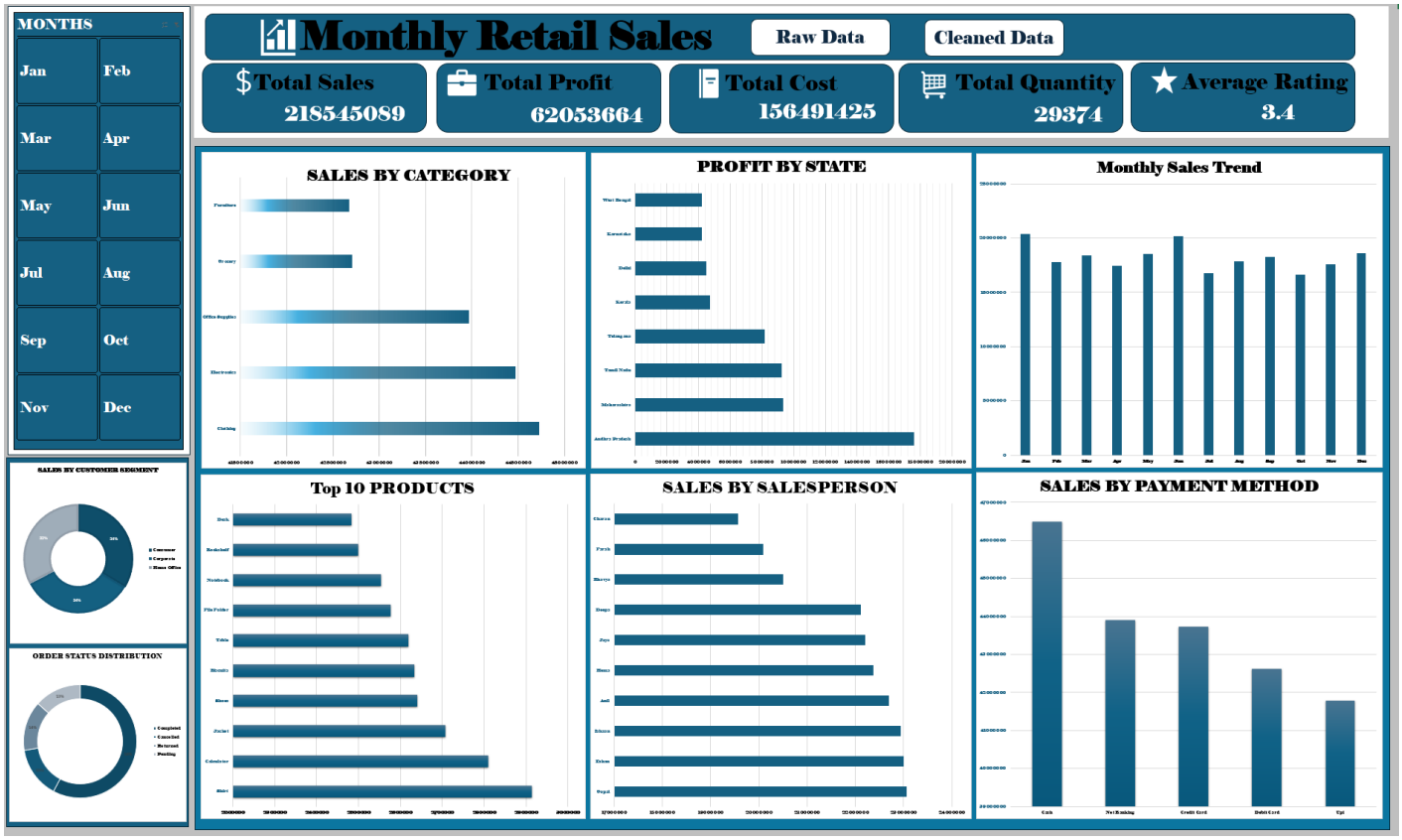


- I added month slicers to my dashboard so users can select a particular month and view the related sales performance.
- The slicers were connected to the relevant PivotTables and charts using Excel's Report Connections feature.
- When I select a month, the connected charts and dashboard values update automatically to show the results for that month.
- This feature makes it easy to compare sales performance across different months without changing the original dataset.
- By using month slicers, I made my dashboard more interactive and user-friendly, helping users explore sales data more easily

## 8.4 Dashboard Layout and Formatting

- I used shapes and text boxes in Excel to create the KPI cards and arrange the dashboard.
- I added suitable titles, chart types, and number formats to make the dashboard clear and easy to understand
- The KPI cards and charts were arranged neatly so users can quickly find important information.
- 4. This helped create a clean and user-friendly dashboard to present retail sales data in a simple way

## 8.5 Final Dashboard




- By combining KPI cards, eight charts, month slicers, and linked cell references, an interactive dashboard was created for retail sales analysis.
- The dashboard displays important information about sales, profit, cost, product performance, and other business metrics in one place.
- Month slicers allow users to filter data and compare sales performance across different months.
- Overall, the dashboard presents retail sales information clearly and makes it easier to understand business performance.

# 9.Conclusion

- ❖ This project helped me learn how to clean and organize retail sales data using Microsoft Excel 2024. The dataset was reduced from **10,031 rows to 9,663 rows** after removing duplicates and blank values.
- ❖ I used PivotTables, Pivot Charts, and Excel formulas to summarize and analyze retail sales data.
- ❖ I created five KPI cards to display **Total Sales, Total Profit, Total Cost, Total Quantity, and Average Rating**.
- ❖ I developed eight charts to analyze sales by category, profit by state, monthly sales trends, customer segments, top 10 products by profit, salesperson performance, order status, and payment method.
- ❖ The analysis showed that **Electronics generated the highest total sales**, with approximately ₹12.15 crore in sales, making it the leading category in the dataset.
- ❖ The charts helped compare monthly sales, identify the most profitable products, evaluate salesperson performance, and understand sales distribution across customer segments and payment methods. These comparisons can help identify areas for further business analysis.
- ❖ I added month slicers to make the dashboard interactive and allow users to compare sales performance across different months.
- ❖ Overall, this project improved my Excel skills and helped me understand how to turn raw retail data into useful business insights through data cleaning, analysis, and dashboard creation.